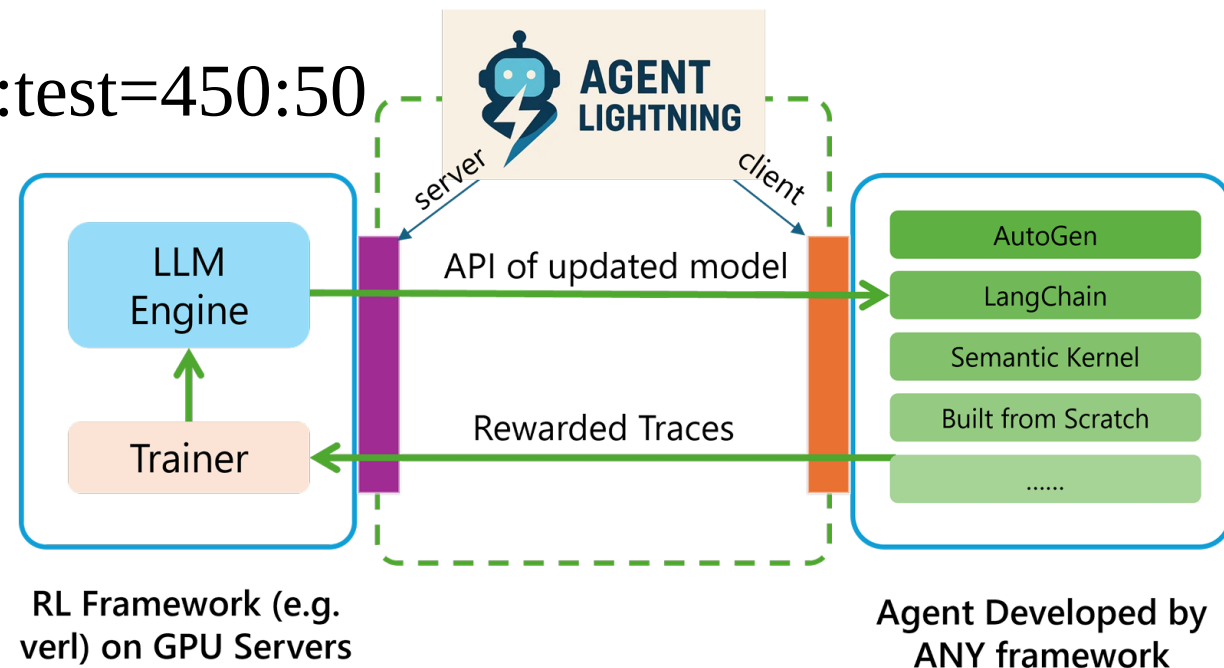


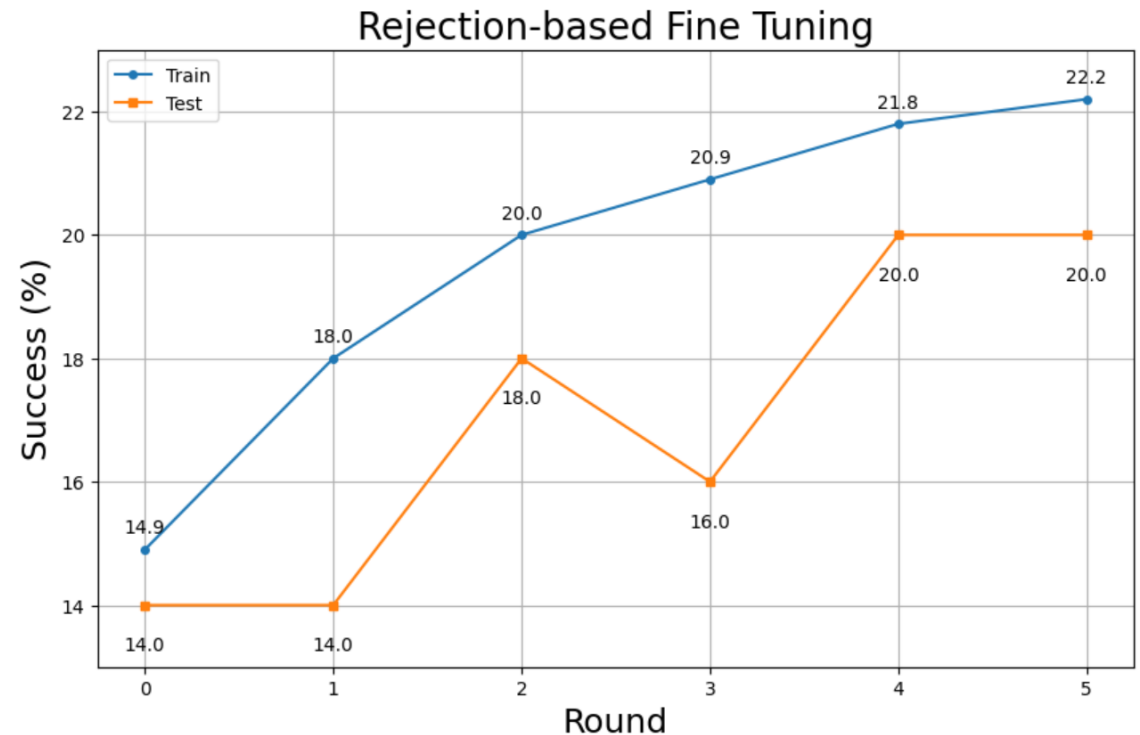
Trajectory Collection

- Middleware wraps vllm server to collect token id and probability
- Claude Code agent side only receives api base url of vllm server
- Qwen3-8b, 8*b200 GPUs
- Sample 8 trajs per instance, t=1.0, disable other sampling params
- Max step per rollout = 50
- Dataset: SWE-bench-Verified, train:test=450:50



Rejection-based Fine-tuning

- Input the whole trajectory, mask loss of prompt and tool-error response
- Duplicate the successful samples of a task instance to 8 to balance data
- $lr=1e-5$, `mini_batch_size=64` trajectories, `batch_size=all` the 450 tasks
- 3 epochs for each round, 5 rounds



Reinforcement Learning

- All steps from 8 sample trajs under a task form one group G
- All steps under the same traj share the same reward=advantage=
$$\frac{r - \overline{r}_G}{Std_G}$$
- DAPO algorithm, clip_low=0.2, clip_high=0.28, disable KL loss
- lr=1e-6, batch_size=mini_batch_size=64 SWE tasks
- 1 epoch for each round, 50 rounds

